

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	139782	46.022705
1	[Git, Change, Create]	45409	14.950745
2	[Git, Change, Delete]	22797	7.505828
3	[Java, ByteCode, Member, Method]	13783	4.538002
4	[Java, ByteCode, Parameter]	13447	4.427375
5	[Git, Commit]	11584	3.813989
6	[File, Git]	8331	2.742951
7	[Java, ByteCode, Bound, ParameterizedType]	7316	2.408766
8	[Java, ByteCode, Bound]	7304	2.404815
9	[Git, Change, Rename]	7018	2.310650
10	[Java, ByteCode, Member, Field]	3704	1.219528
11	[Java, Value, ByteCode, Annotation]	2955	0.972923
12	[Java, ByteCode, Bound, WildcardType]	2950	0.971277
13	[Xml, Element]	2162	0.711830
14	[Java, ByteCode, Member, Method, Constructor]	2160	0.711172
15	[Xml, Text]	1450	0.477407
16	[Java, ByteCode, TypeVariable, Bound]	1112	0.366122
17	[Java, ByteCode, Member, Method, Lambda]	986	0.324637
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	903	0.297309
19	[Type, File, Java, ByteCode, Class]	868	0.285786
20	[Json, Key]	739	0.243313
21	[Value, Json, Scalar]	723	0.238045
22	[Java, Value, ByteCode, Primitive]	685	0.225534
23	[Type, File, Java, ByteCode, JavaType]	659	0.216973
24	[Java, ByteCode, GenericDeclaration, Member, M...	579	0.190634
25	[Type, File, Java, ByteCode, ExternalType]	419	0.137954
26	[Git, Change, Copy]	298	0.098115
27	[Author, Git, Person]	297	0.097786
28	[Value, Array]	293	0.096469
29	[Committer, Git, Person]	242	0.079678

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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46.02% (139782)



Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000329
1	[Git, Branch]	1	0.000329
2	[Repository, File, Git]	1	0.000329
3	[File, Json]	2	0.000658
4	[File]	3	0.000988
5	[Java, ByteCode, GenericDeclaration, Member, M...	4	0.001317
6	[Maven, Exclusion]	5	0.001646
7	[Value, Array, Json]	6	0.001975
8	[Type, File, Java, ByteCode, Void]	9	0.002963
9	[File, Maven, Xml, Pom, Document]	9	0.002963
10	[Java, ManifestSection]	9	0.002963
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002963
12	[File, Java, Manifest]	9	0.002963
13	[File, Java, ServiceLoader]	11	0.003622
14	[File, Java, Properties]	12	0.003951
15	[Maven, ExecutionGoal]	16	0.005268
16	[Maven, PluginExecution]	16	0.005268
17	[Xml, Attribute]	18	0.005926
18	[jQAssistant, Rule, Concept]	19	0.006256
19	[Maven, Plugin]	21	0.006914
20	[Maven, Configuration]	21	0.006914
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006914
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007243
23	[Type, File, Java, ByteCode, Enum]	30	0.009877
24	[Type, File, Java, ByteCode, PrimitiveType]	31	0.010207
25	[Xml, Namespace]	36	0.011853
26	[Type, File, Java, ByteCode, Annotation]	44	0.014487
27	[File, Directory]	45	0.014816
28	[Type, File, Java, ByteCode, Class, Throwable]	56	0.018438
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.027986

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

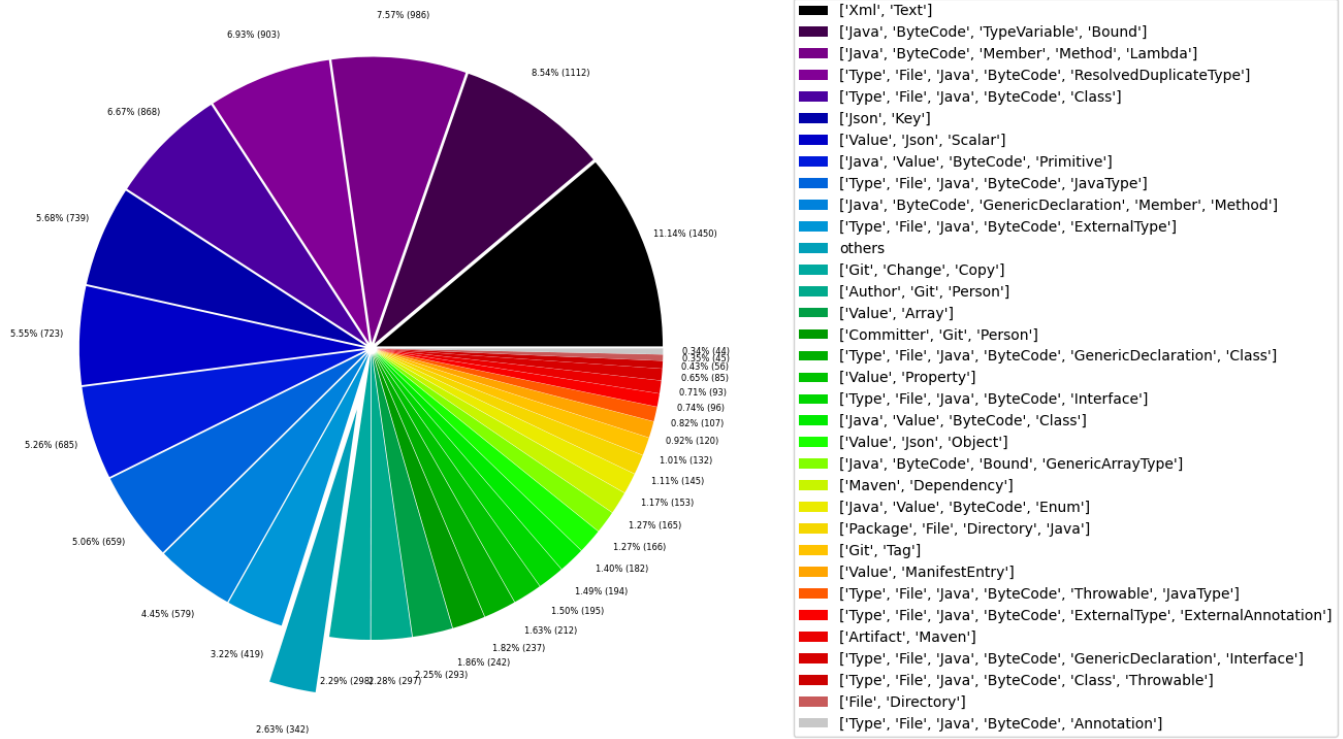


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

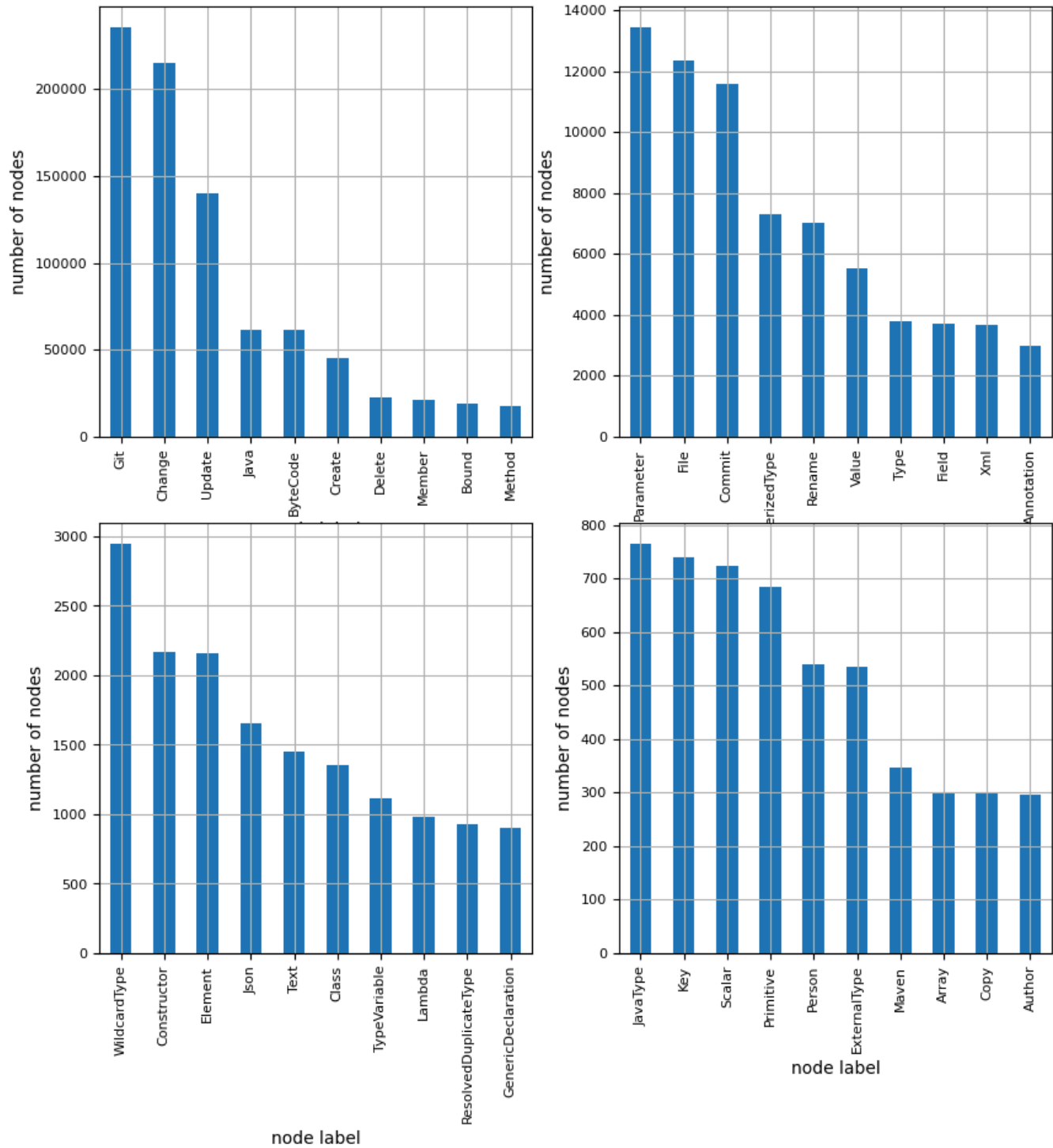
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	235892	77.666566
1	Change	215304	70.888043
2	Update	139782	46.022705
3	Java	61475	20.240416
4	ByteCode	61280	20.176213
5	Create	45409	14.950745
6	Delete	22797	7.505828
7	Member	21216	6.985289
8	Bound	18848	6.205634
9	Method	17512	5.765761
10	Parameter	13447	4.427375
11	File	12359	4.069155
12	Commit	11584	3.813989
13	ParameterizedType	7316	2.408766
14	Rename	7018	2.310650
15	Value	5523	1.818427
16	Type	3782	1.245209
17	Field	3704	1.219528
18	Xml	3675	1.209980
19	Annotation	2999	0.987410
20	WildcardType	2950	0.971277
21	Constructor	2164	0.712489
22	Element	2162	0.711830
23	Json	1652	0.543915
24	Text	1450	0.477407
25	Class	1355	0.446129
26	TypeVariable	1112	0.366122
27	Lambda	986	0.324637
28	ResolvedDuplicateType	925	0.304553
29	GenericDeclaration	905	0.297968
30	JavaType	766	0.252203
31	Key	739	0.243313
32	Scalar	723	0.238045
33	Primitive	685	0.225534
34	Person	539	0.177464
35	ExternalType	536	0.176476
36	Maven	346	0.113919
37	Array	299	0.098445
38	Copy	298	0.098115
39	Author	297	0.097786

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 967956

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	215304	22.243160
1	MODIFIES	215304	22.243160
2	UPDATES	139782	14.440946
3	CREATES	52725	5.447045
4	INVOKES	37734	3.898318
5	DELETES	29815	3.080202
6	CHANGED_TOGETHER_WITH	26193	2.706011
7	COMMITTED	23168	2.393497
8	DEPENDS_ON	22889	2.364674
9	OF_TYPE	22204	2.293906
10	DECLARES	21681	2.239875
11	OF_RAW_TYPE	17460	1.803801
12	HAS	14602	1.508540
13	HAS_PARENT	14170	1.463910
14	RETURNS	13073	1.350578
15	HAS_COMMIT	11584	1.196749
16	READS	9673	0.999322
17	HAS_ACTUAL_TYPE_ARGUMENT	8445	0.872457
18	HAS_FILE	8331	0.860680
19	RENAMES	7018	0.725033
20	OF_GENERIC_TYPE	6036	0.623582
21	RESOLVES_TO	4282	0.442375
22	SIMILAR	4130	0.426672
23	WRITES	4042	0.417581
24	CONTAINS	4005	0.413758
25	HAS_NEW_NAME	3662	0.378323
26	RETURNS_GENERIC	3616	0.373571
27	ANNOTATED_BY	2943	0.304043
28	REQUIRES	2267	0.234205
29	HAS_FIRST_CHILD	2162	0.223357

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

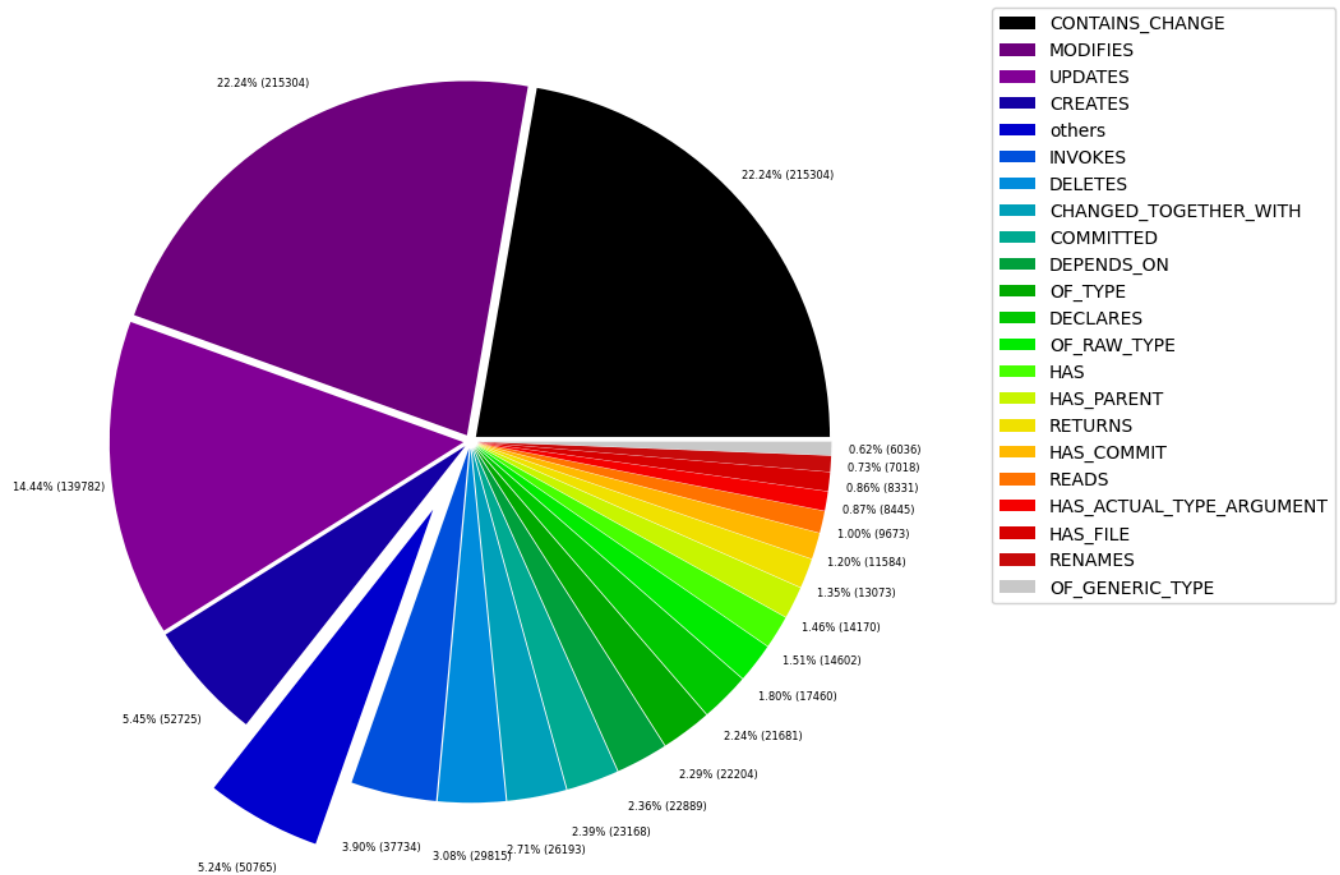


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

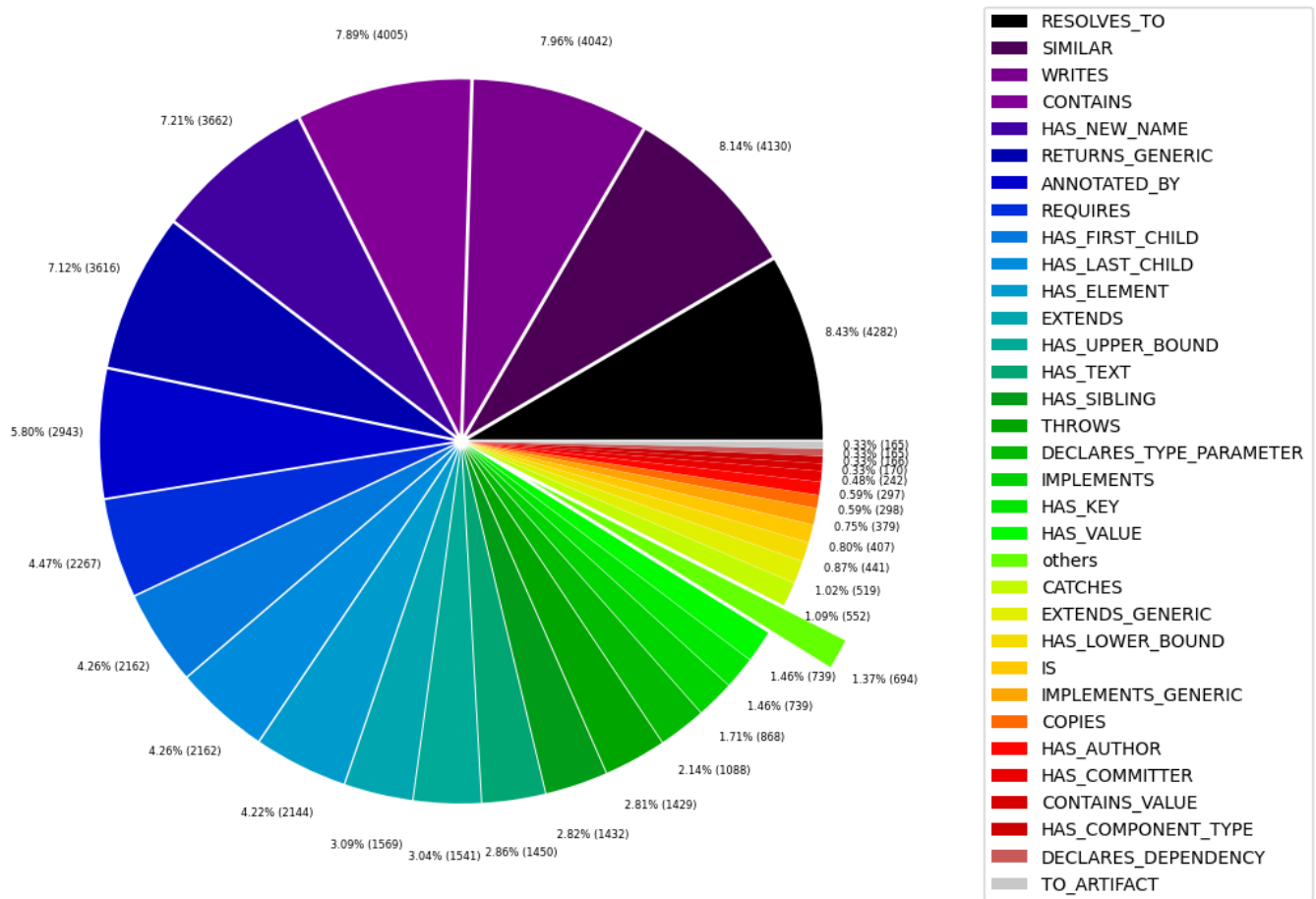
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000103
1	HAS_BRANCH	1	0.000103
2	HAS_HEAD	2	0.000207
3	THROWS_GENERIC	5	0.000517
4	EXCLUDES	5	0.000517
5	DESCRIBES	9	0.000930
6	HAS_ROOT_ELEMENT	11	0.001136
7	HAS_GOAL	16	0.001653
8	HAS_EXECUTION	16	0.001653
9	OF_NAMESPACE	18	0.001860
10	HAS_ATTRIBUTE	18	0.001860
11	INCLUDES_CONCEPT	19	0.001963
12	USES_PLUGIN	21	0.002170
13	IS_ARTIFACT	21	0.002170
14	HAS_CONFIGURATION	21	0.002170
15	REQUIRES_TYPE_PARAMETER	24	0.002479
16	REQUIRES_CONCEPT	28	0.002893
17	DECLARES_NAMESPACE	36	0.003719
18	HAS_DEFAULT	39	0.004029
19	COPY_OF	119	0.012294
20	HAS_TAG	132	0.013637
21	ON_COMMIT	132	0.013637
22	DECLARES_DEPENDENCY	165	0.017046
23	TO_ARTIFACT	165	0.017046
24	HAS_COMPONENT_TYPE	166	0.017150
25	CONTAINS_VALUE	170	0.017563
26	HAS_COMMITTER	242	0.025001
27	HAS_AUTHOR	297	0.030683
28	COPIES	298	0.030787
29	IMPLEMENTS_GENERIC	379	0.039155

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	139782	11584	
1	[Git, Update, Change]	MODIFIES	[File, Git]	139782	139782	
2	[Git, Update, Change]	UPDATES	[File, Git]	139782	139782	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	45409	11584	
4	[Git, Change, Create]	CREATES	[File, Git]	45409	45409	
5	[Git, Change, Create]	MODIFIES	[File, Git]	45409	45409	
6	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	23198	13687	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22797	11584	
8	[Git, Change, Delete]	DELETES	[File, Git]	22797	22797	
9	[Git, Change, Delete]	MODIFIES	[File, Git]	22797	22797	
10	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	20059	8331	
11	[Git, Commit]	HAS_PARENT	[Git, Commit]	14161	11584	
12	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11584	1	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	11584	297	
14	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11584	242	
15	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8706	13687	
16	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8599	13687	
17	[Repository, File, Git]	HAS_FILE	[File, Git]	8331	1	
18	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11584	
19	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	6235	13447	
24	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8331	
25	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3488	7304	
26	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3144	7316	
27	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7316	
28	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2709	13447	
29	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2577	2160	

Graph Density

```
total_number_of_nodes (vertices): 303724
total_number_of_relationships (edges): 967956
-> total directed graph density: 1.0492979500958668e-05
-> total directed graph density in percent: 0.0010492979500958668
```